

PROFILE SUMMARY

- AI Engineer with 6 years of experience designing and shipping LLM-powered products across answer engines, customer-support agents, and developer tools, specializing in prompt engineering, retrieval-augmented generation, and agent orchestration.
- Solid technical background across LLMs (Claude, GPT-5), agent frameworks (LangChain, LangGraph), vector databases (Pinecone, pgvector), LLM observability (LangSmith, Langfuse), and languages (Python, TypeScript) with strong fundamentals in REST APIs, function calling, and structured outputs.
- Deep expertise in end-to-end LLM applications, agentic workflows, retrieval-augmented generation, and structured-output orchestration, leveraging methodologies such as prompt iteration loops and LLM-as-judge evaluation to drive reliable, cost-aware, and observable AI products.
- Engaged collaborator working cross-functionally with Product, Design, and domain experts in Agile environments, contributing to AI use-case discovery, eval design, and post-launch retrospectives with a pragmatic, ownership-first mindset.
- Emerging leader who shares technical excellence and fosters a culture of evaluation-first thinking and responsible-AI discipline through PR reviews and runbooks, while leading AI guild sessions and authoring widely adopted prompt-and-eval templates.

TECHNICAL SKILLS

LLMs & Foundation Models:	Claude, GPT-5, Gemini, Llama, Mistral
Agent & Orchestration:	LangChain, LangGraph, CrewAI, Pydantic AI, MCP
RAG & Retrieval:	Pinecone, Weaviate, Chroma, pgvector, Cohere reranker, hybrid search
Languages & Scripting:	Python, TypeScript, SQL, Bash
LLM Observability:	LangSmith, Langfuse, Arize, OpenTelemetry, Helicone
Evaluation & QA:	LLM-as-judge, golden datasets, RAGAS, DeepEval, human-in-the-loop
Safety & Guardrails:	Input/output filtering, PII redaction, jailbreak resistance, OWASP LLM Top 10
Cloud & Inference:	AWS Bedrock, Azure OpenAI, GCP Vertex AI, prompt caching, model routing

EDUCATION

University of Washington B.S. in Computer Science Seattle, WA • Sep 2016 - Jun 2020

WORK EXPERIENCE

Perplexity Senior AI Engineer San Francisco, CA • Sep 2022 - Present

- Owned the end-to-end LLM application layer for Perplexity Pro Search serving 5M+ paid subscribers, leading design across prompt orchestration, retrieval pipelines, and agentic search workflows across 18 product features in a polyglot Python and TypeScript environment.
- Designed the prompt-engineering framework for citation-grounded answers, anchored on few-shot prompting with structured-output JSON schemas and multi-step prompt chaining across 4 model providers; lifted answer accuracy from 71% to 89% on the internal eval set.
- Built the retrieval-augmented generation pipeline on Pinecone and pgvector around hybrid BM25 + semantic search with adaptive chunking by content type and Cohere reranking, indexing 180M+ documents at 240ms p95 retrieval latency.
- Architected the multi-agent search system in LangGraph with planner + executor + critic roles, MCP for tool use against

live data, and graceful fallback chains, handling **2.4M** agentic queries per day at **96%** successful task completion.

- Stood up the team's LLM evaluation framework around **LLM-as-judge graders**, human-in-the-loop golden sets, and regression suites in **LangSmith**, running **40+** structured evals that gated **12** model launches and caught **9** pre-prod hallucination regressions.
- Implemented the AI safety and guardrail layer with **input and output filtering**, jailbreak resistance via constitutional rules, content-moderation routing, and **prompt-boundary redaction** of PII; cut policy-violating output rate from **3.2%** to **0.18%** across production traffic.
- Optimized inference cost through **prompt caching**, semantic caching of repeat queries, token-budget streaming, and **dynamic model routing** (Sonnet for routine, Opus for complex), cutting per-query cost by **62%** (~\$1.4M annual run-rate) without quality regression.

Intercom **AI Engineer**

San Francisco, CA • Aug 2020 - Aug 2022

- Owned model selection and fine-tuning for the **Fin support agent**, evaluating **Claude vs GPT-4 vs Llama** and applying **LoRA fine-tuning** on **18k** labeled support tickets; shipped a custom adapter that beat the closed-source baseline on resolution rate (+8%) at **40%** lower cost-per-conversation.
- Built the AI observability stack on **Langfuse** with **OpenTelemetry** traces and **prompt versioning**, A/B test routing, and drift-monitoring dashboards covering **22** production prompts; surfaced **15** silent regressions in the first six months.
- Built Fin's structured-output orchestration layer in **LangChain** with Pydantic schemas, integrating **9** internal tools (CRM lookup, billing API, knowledge-base search) via function calling, powering **800k+** resolved conversations per month at **sub-second p50** latency.
- Worked closely with **Product, Design, and Domain experts** across **5** product surfaces to negotiate **AI use-case prioritization**, non-determinism UX patterns, and launch quality bars; authored **8** responsible-AI RFCs that shaped the org's launch playbook and onboarded **9** new AI engineers.